Final Project: Supervised Classification of Sentinel-2 Imagery

A Comparative Analysis of Machine Learning Architectures for Satellite-Based Surface Monitoring in Hunga Tonga–Hunga Ha'apai

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May 2026

Abstract

This final project presents a supervised machine learning workflow for the classification of Sentinel-2 multispectral imagery over the Hunga Tonga–Hunga Ha'apai volcanic system. The project focused on the construction of a pixel-level training dataset, the comparison of five machine learning architectures, and the generation of thematic raster outputs for surface and atmospheric class mapping. The evaluated models were Decision Tree, Support Vector Machine, Artificial Neural Network, K-Nearest Neighbor, and Naive Bayes.

An initial classification was developed using five classes: ocean water, bare soil, vegetation, cloud, and cloud shadow. The best-performing model was K-Nearest Neighbor, with an accuracy of 0.9967, a Kappa coefficient of 0.9959, and a macro F1-score of 0.9967. After visual inspection, the methodology was improved by defining a cropped region of interest around the island and adding a sixth class, Dark_ocean_water, to better separate deep or low-reflectance ocean pixels from cloud shadows. In this improved version, K-Nearest Neighbor again achieved the best performance, with an accuracy of 0.9925, a Kappa coefficient of 0.9910, and a macro F1-score of 0.9925. The final outputs include classified GeoTIFF rasters, QGIS visualizations, confusion matrices, model comparison plots, and area quantification tables for both the cropped region and the complete Sentinel-2 scene.

1 Introduction

Satellite remote sensing provides an effective way to monitor surface changes, land-cover patterns, and environmental conditions in areas that are difficult to access directly. Multispectral sensors such as Sentinel-2 capture reflectance information in several spectral bands, allowing the identification of different surface materials and atmospheric elements through their spectral responses.

This project focuses on the Hunga Tonga–Hunga Ha'apai volcanic system, located in the South Pacific. The selected Sentinel-2 image corresponds to November 18, 2021, before the major volcanic eruption that occurred in January 2022. This date was selected because the island was still visible and the scene contained different spectral elements relevant for supervised classification, including ocean water, exposed volcanic land, vegetation, clouds, and cloud shadows.

The objective of this project was to implement a complete supervised classification workflow using Sentinel-2 imagery and compare the performance of five machine learning models. The project included image preprocessing, training polygon digitization, pixel-level dataset construction, hyperparameter optimization, model evaluation, raster inference, map generation, and area quantification by class.

2 Study Area and Case Selection

The study area corresponds to the Hunga Tonga–Hunga Ha'apai volcanic system. This location is relevant for satellite-based environmental monitoring because of the significant geomorphological changes associated with volcanic activity. The selected image represents the pre-eruption condition of the island, providing a useful baseline for surface classification.

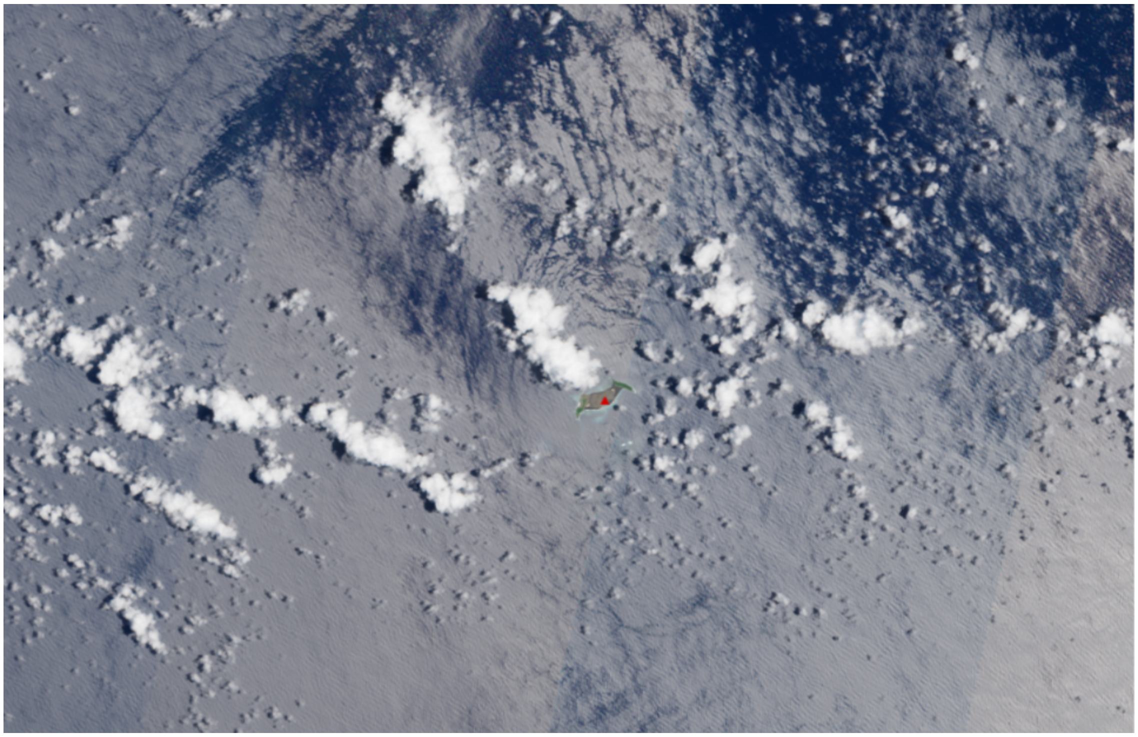

Figure 1: Location of the Hunga Tonga–Hunga Ha'apai study area.

The selected image contains both surface and atmospheric features. Therefore, the classification task was not limited to land-cover categories but also included clouds and cloud shadows, which are important elements in optical satellite image interpretation.

3 Data Source and Spectral Bands

The project used Sentinel-2 multispectral imagery. Ten spectral bands were selected as predictor variables. Four of them are available at 10 m spatial resolution, while six are available at 20 m spatial resolution. The 20 m bands were resampled to the 10 m grid to ensure pixel-level alignment during training and inference.

Table 1: Sentinel-2 bands used in the classification workflow.

Band	**Description**	**Original resolution**
B02	Blue	10 m
B03	Green	10 m
B04	Red	10 m
B05	Red Edge 1	20 m
B06	Red Edge 2	20 m
B07	Red Edge 3	20 m
B08	Near Infrared	10 m
B8A	Narrow Near Infrared	20 m
B11	Short-Wave Infrared 1	20 m
B12	Short-Wave Infrared 2	20 m

Two visual compositions were used in QGIS to support class interpretation. A natural color composite was created using B04, B03, and B02, while a false color composite was created using B08, B04, and B03. These visualizations helped identify vegetation, exposed soil, ocean water, clouds, and shadow areas.

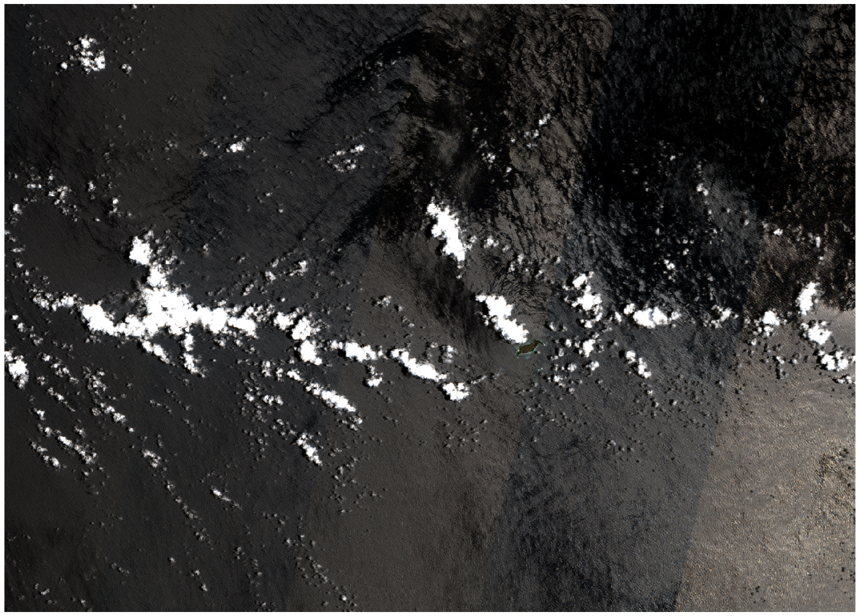

Figure 2: Natural color Sentinel-2 composite of the study area.

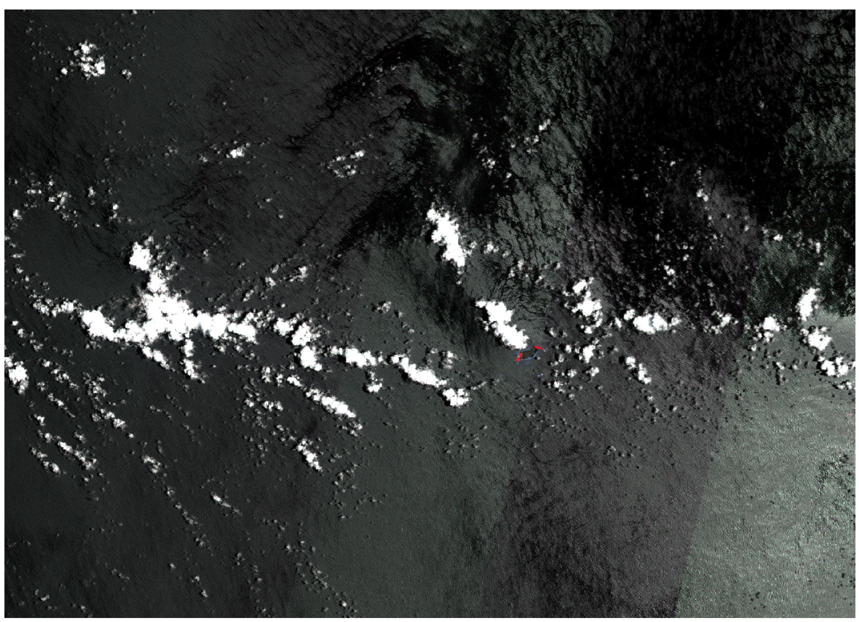

Figure 3: False color Sentinel-2 composite used for visual interpretation.

4 General Methodology

The complete workflow was divided into three main stages:

1. Initial supervised classification of the complete Sentinel-2 scene using five classes.
2. Improved classification over a cropped region of interest using six classes.
3. Application of the improved model to the complete Sentinel-2 scene.

The general methodological process included:

1. Selection of Sentinel-2 bands.
2. Creation of training polygons in QGIS.
3. Extraction of pixel-level spectral values from the polygons.
4. Construction of tab-separated value datasets.
5. Dataset balancing.
6. Model training and hyperparameter optimization.
7. Model evaluation using standard classification metrics.
8. Selection of the best-performing model.
9. Raster inference over the image.
10. Export of thematic GeoTIFF files.
11. Visualization in QGIS using discrete class symbology.
12. Area quantification by class.

5 Initial Classification with Five Classes

5.1 Initial Class Definition

The first version of the classification workflow used five classes. These classes were selected based on visible features in the Sentinel-2 image and their relevance for the case study.

Table 2: Initial classification classes.

Class ID	**Class name**	**Description**
1	Ocean_water	Open ocean water surrounding the island.
2	Bare_soil	Exposed volcanic soil, sand, or non-vegetated land.
3	Vegetation	Vegetated areas identified using natural and false color interpretation.
4	Cloud	Bright cloud features visible in the optical image.
5	Cloud_shadow	Dark shadow areas associated mainly with cloud cover.

5.2 Training Polygons

The initial training polygons were created in QGIS over visually homogeneous areas. Mixed areas such as coastlines, cloud edges, and transitional zones were avoided to reduce spectral ambiguity.

Table 3: Initial number of training polygons by class.

Class name	**Number of polygons**
Ocean_water	17
Cloud	14
Bare_soil	14
Vegetation	14
Cloud_shadow	14

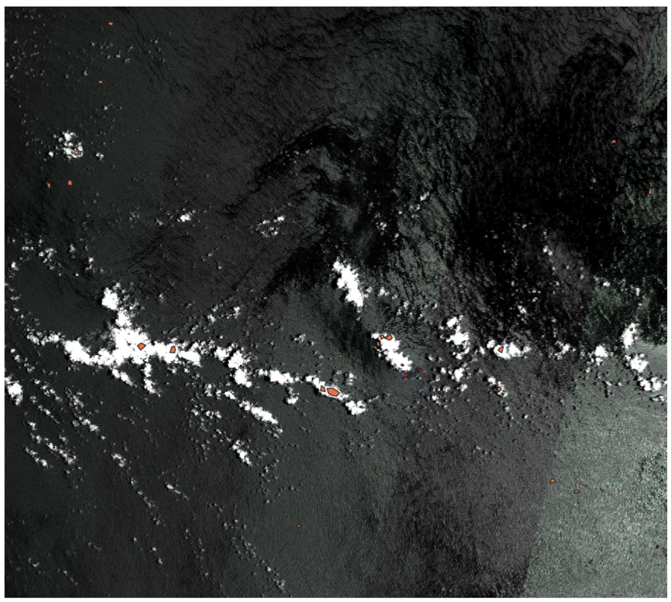

Figure 4: Initial training polygons digitized over the Sentinel-2 image.

5.3 Initial TSV Dataset

The training polygons were used to extract pixel-level spectral signatures from the ten Sentinel-2 bands. Each row in the dataset represented a pixel located inside a training polygon. The resulting file was named:

`training_data.tsv`

Table 4: Structure of the exported TSV dataset.

Column	Description
Latitude	Latitude of the pixel center.
Longitude	Longitude of the pixel center.
B02	Blue band value.
B03	Green band value.
B04	Red band value.
B05	Red Edge 1 band value.
B06	Red Edge 2 band value.
B07	Red Edge 3 band value.
B08	Near Infrared band value.
B8A	Narrow Near Infrared band value.
B11	Short-Wave Infrared 1 band value.
B12	Short-Wave Infrared 2 band value.
class_id	Numerical class identifier.
class_name	Categorical class label.

The initial extraction produced 131,840 pixel samples.

Table 5: Original distribution of extracted pixel samples in the initial dataset.

Class name	Extracted pixels
Cloud	63,930
Ocean_water	50,572
Cloud_shadow	11,861
Bare_soil	3,021
Vegetation	2,456
Total	**131,840**

Since the dataset was imbalanced, it was balanced using the minority class count. Vegetation had 2,456 samples, so 2,456 samples were randomly selected from each class.

Table 6: Balanced initial dataset.

Class name	Samples used
Bare_soil	2,456
Cloud	2,456
Cloud_shadow	2,456
Ocean_water	2,456
Vegetation	2,456
Total	**12,280**

The balanced dataset was divided using a stratified train-test split.

Table 7: Initial training and testing split.

Subset	Number of samples
Training set	9,210
Testing set	3,070

6 Machine Learning Models

Five supervised machine learning models were implemented and compared:

1. **Decision Tree:** a rule-based classifier that separates classes using hierarchical decision rules.
2. **Support Vector Machine:** a margin-based model that separates classes using hyperplanes in the feature space.
3. **Artificial Neural Network:** a multilayer model capable of learning nonlinear relationships between spectral bands and classes.
4. **K-Nearest Neighbor:** a non-parametric classifier that assigns each sample to the most common class among its nearest neighbors.
5. **Naive Bayes:** a probabilistic classifier based on Bayes' theorem and an independence assumption among features.

7 Hyperparameter Optimization and Evaluation Metrics

A hyperparameter optimization process was performed using grid search and cross-validation. The objective was to select the best configuration for each model based on macro F1-score.

Table 8: Best hyperparameters obtained for each model.

Model	Best hyperparameters
Decision Tree	criterion = entropy; max_depth = 20; min_samples_split = 5
SVM	C = 10; gamma = scale; kernel = linear
ANN	activation = tanh; hidden_layer_sizes = (100,); learning_rate_init = 0.01
KNN	metric = euclidean; n_neighbors = 3; weights = distance
Naive Bayes	var_smoothing = 1e-06

The models were evaluated using the following metrics:

- **Accuracy:** proportion of correctly classified samples.
- **Kappa coefficient:** agreement between predictions and reference labels beyond chance.
- **Precision:** proportion of predicted samples that were correct.
- **Recall:** proportion of actual samples correctly identified.
- **F1-score:** harmonic mean between precision and recall.

Macro-averaged metrics were used because they assign equal importance to each class.

8 Initial Model Comparison Results

The initial model comparison showed that KNN achieved the best performance among the five architectures.

Table 9: Initial model comparison using the balanced test set.

Model	Accuracy	Kappa	Precision	Recall	F1-score
KNN	0.9967	0.9959	0.9968	0.9967	0.9967
Decision Tree	0.9941	0.9927	0.9941	0.9941	0.9941
SVM	0.9853	0.9817	0.9862	0.9853	0.9853
ANN	0.9847	0.9809	0.9855	0.9847	0.9847
Naive Bayes	0.8814	0.8518	0.8785	0.8814	0.8794

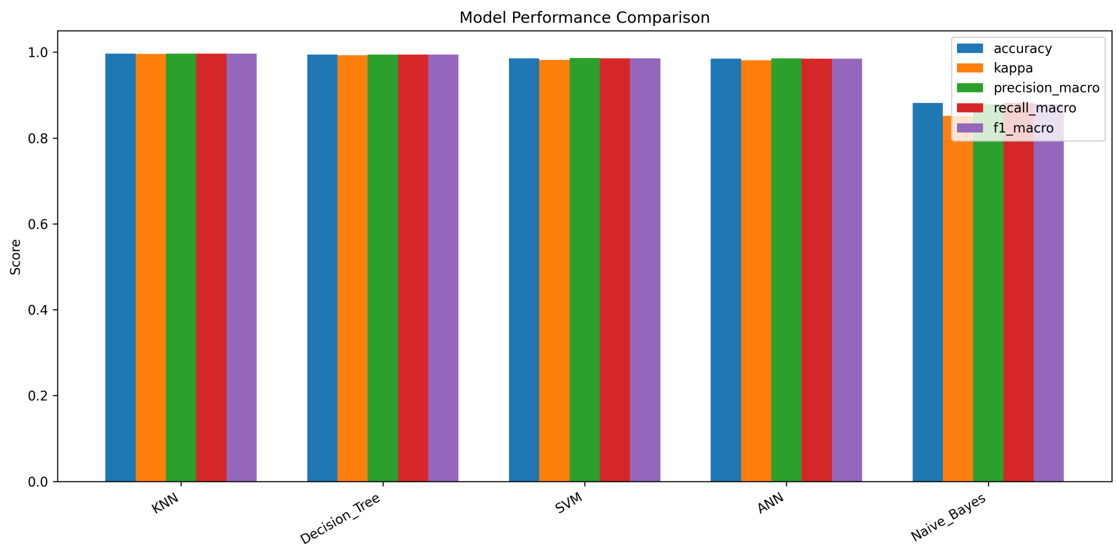

Figure 5: Initial comparison of model performance metrics.

The best hyperparameters obtained in the initial classification were:

Table 10: Initial best hyperparameters obtained for each model.

Model	Best hyperparameters
Decision Tree	criterion = entropy; max_depth = 20; min_samples_split = 5
SVM	C = 10; gamma = scale; kernel = linear
ANN	activation = tanh; hidden_layer_sizes = (100,); learning_rate_init = 0.01
KNN	metric = euclidean; n_neighbors = 3; weights = distance
Naive Bayes	var_smoothing = 1e-06

9 Initial Confusion Matrix

Since KNN achieved the highest performance in the initial workflow, its confusion matrix was analyzed in greater detail.

Table 11: Initial confusion matrix for the KNN model.

Actual / Predicted	Bare_soil	Cloud	Cloud_shadow	Ocean_water	Vegetation
Bare_soil	611	0	0	3	0
Cloud	0	614	0	0	0
Cloud_shadow	0	0	613	1	0
Ocean_water	0	0	6	608	0
Vegetation	0	0	0	0	614

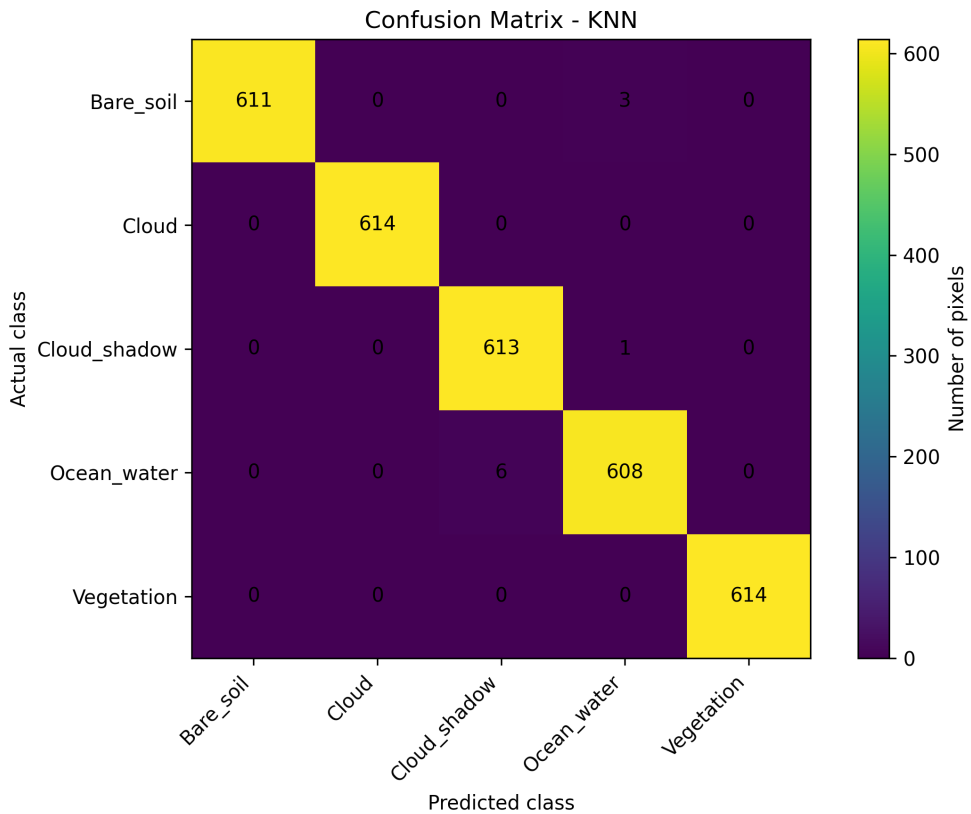

Figure 6: Initial confusion matrix for the KNN classifier.

The confusion matrix shows that most errors occurred between Ocean_water and Cloud_shadow. This result was expected because both classes can show low reflectance values in visible bands. This observation motivated a methodological improvement in the second version of the classification workflow.

10 Initial Full-Scene Raster Classification

The best initial model was applied to the complete Sentinel-2 scene to generate a thematic raster. The output was exported as a GeoTIFF and visualized in QGIS using discrete symbology.

Table 12: Initial area by class over the complete Sentinel-2 scene.

Class name	Pixel count	Area ha	Area km^2
Ocean_water	113,631,591	1,136,315.91	11,363.1591
Cloud_shadow	4,142,218	41,422.18	414.2218
Cloud	1,951,958	19,519.58	195.1958
Bare_soil	827,967	8,279.67	82.7967
Vegetation	6,661	66.61	0.6661

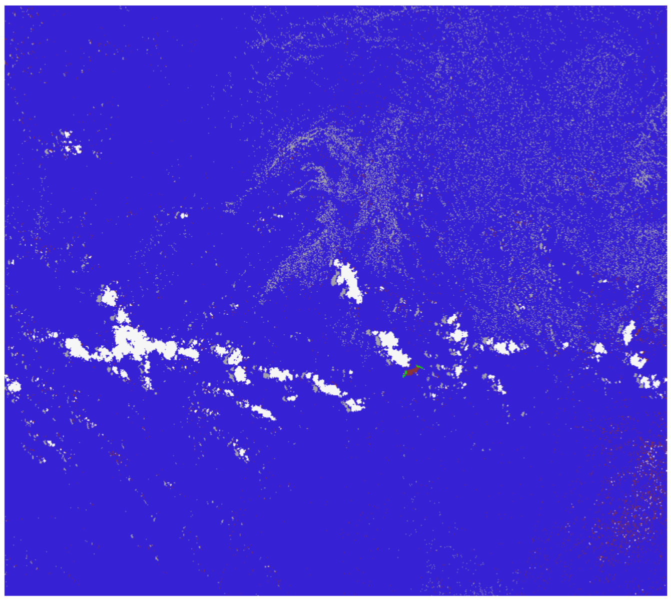

Figure 7: Initial classified area by class over the complete Sentinel-2 scene.

11 Improved Classification Workflow

11.1 Motivation for Improvement

After the initial classification, a visual review showed that some dark ocean areas could be confused with cloud shadows. This is a common issue in optical satellite imagery because deep water, rough sea texture, low illumination, and cloud shadows may all produce dark spectral responses.

To improve the classification, two methodological changes were introduced:

1. A cropped region of interest was created around the island.
2. A new class named Dark_ocean_water was added.

This improved workflow allowed the model to distinguish between normal ocean water, dark ocean water, and actual cloud shadow areas.

11.2 Region of Interest

A region of interest was digitized in QGIS around the island and exported as a vector file. The original Sentinel-2 bands were then clipped to this region using Python. The clipped 10 m bands had a raster size of 4,346 columns by 2,434 rows.

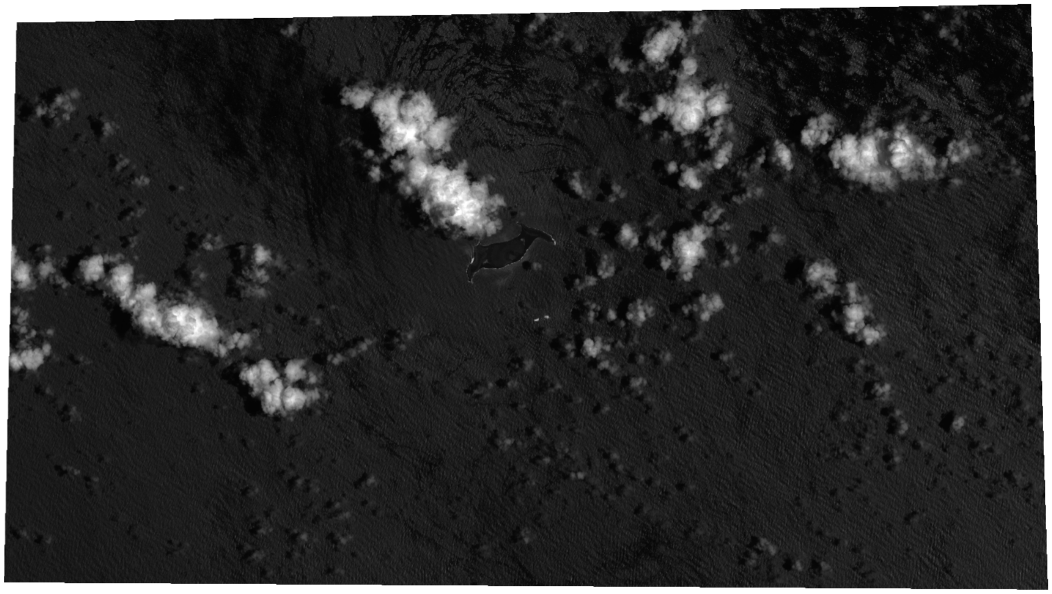

Figure 8: Cropped region of interest used for the improved classification workflow.

11.3 Improved Class Definition

The improved classification workflow used six classes.

Table 13: Improved classification classes.

Class ID	Class name	Description
1	Ocean_water	Open ocean water with typical spectral response.
2	Dark_ocean_water	Dark ocean water related to deep water, wave texture, or low reflectance zones.
3	Bare_soil	Exposed volcanic soil, sand, or non-vegetated land.
4	Vegetation	Vegetated areas on the island.
5	Cloud	Bright cloud features.
6	Cloud_shadow	Dark areas associated with cloud shadows.

11.4 Improved Training Polygons

New training polygons were created for the six-class version. The polygons were digitized over homogeneous zones inside the cropped region.

Table 14: Number of training polygons in the improved workflow.

Class name	Number of polygons
Bare_soil	15
Cloud	15
Ocean_water	14
Dark_ocean_water	14
Vegetation	14
Cloud_shadow	14

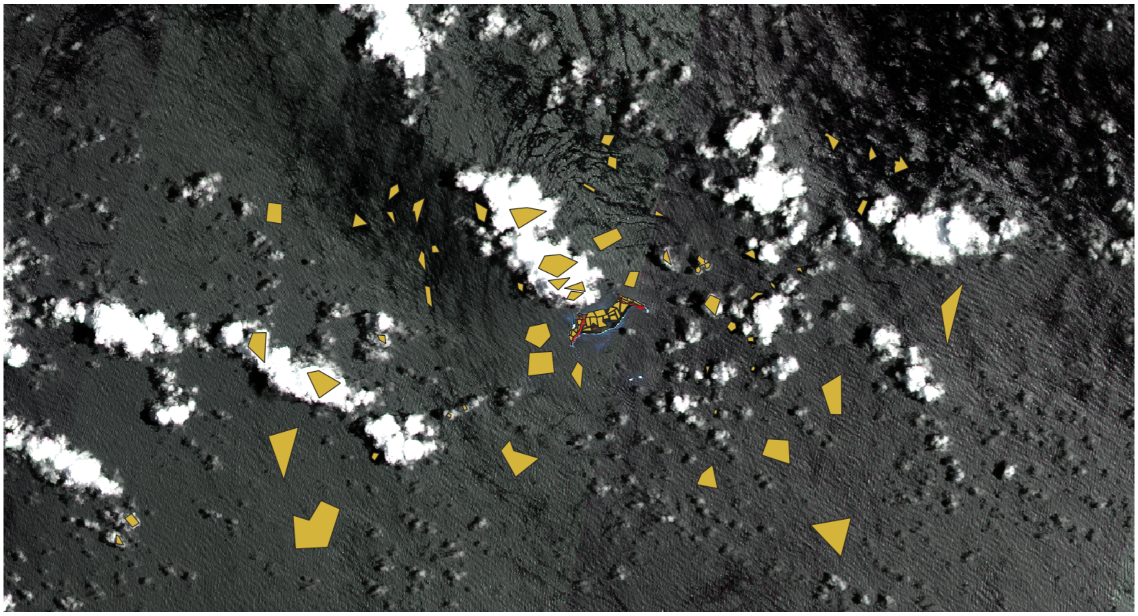

Figure 9: Improved training polygons with six classes.

12 Improved TSV Dataset

The improved training polygons were used to extract pixel-level spectral signatures from the cropped Sentinel-2 bands. The resulting dataset was named:

`training_data_v2.tsv`

The extraction produced 265,651 pixel samples.

Table 15: Original distribution of extracted pixel samples in the improved dataset.

Class name	**Extracted pixels**
Ocean_water	160,570
Cloud	52,064
Dark_ocean_water	25,652
Cloud_shadow	12,311
Bare_soil	11,134
Vegetation	3,920
Total	**265,651**

Because Vegetation was the minority class with 3,920 samples, the dataset was balanced to 3,920 samples per class.

Table 16: Balanced improved dataset.

Class name	**Samples used**
Bare_soil	3,920
Cloud	3,920
Cloud_shadow	3,920
Dark_ocean_water	3,920
Ocean_water	3,920
Vegetation	3,920
Total	**23,520**

The balanced dataset was divided into training and testing subsets using stratified sampling. Compared with the initial balanced dataset of 12,280 samples, the improved balanced dataset contained 23,520 samples, representing a substantial increase in the amount of data used for model training and testing.

Table 17: Improved training and testing split.

Subset	**Number of samples**
Training set	17,640
Testing set	5,880

Table 18: Comparison of balanced datasets between the initial and improved classification phases.

Comparison	**Initial phase**	**Improved phase**	**Increase**
Balanced dataset	12,280	23,520	+11,240 samples
Percentage increase	–	–	+91.53%

13 Improved Model Comparison Results

The five machine learning architectures were trained again using the improved six-class dataset. KNN again achieved the best performance.

Table 19: Improved model comparison using the six-class balanced test set.

Model	**Accuracy**	**Kappa**	**Precision**	**Recall**	**F1-score**
KNN	0.9925	0.9910	0.9926	0.9925	0.9925
ANN	0.9840	0.9808	0.9841	0.9840	0.9840
Decision Tree	0.9833	0.9800	0.9834	0.9833	0.9833
SVM	0.9786	0.9743	0.9790	0.9786	0.9786
Naive Bayes	0.8440	0.8129	0.8473	0.8440	0.8401

Figure 10: Improved comparison of model performance metrics using six classes.

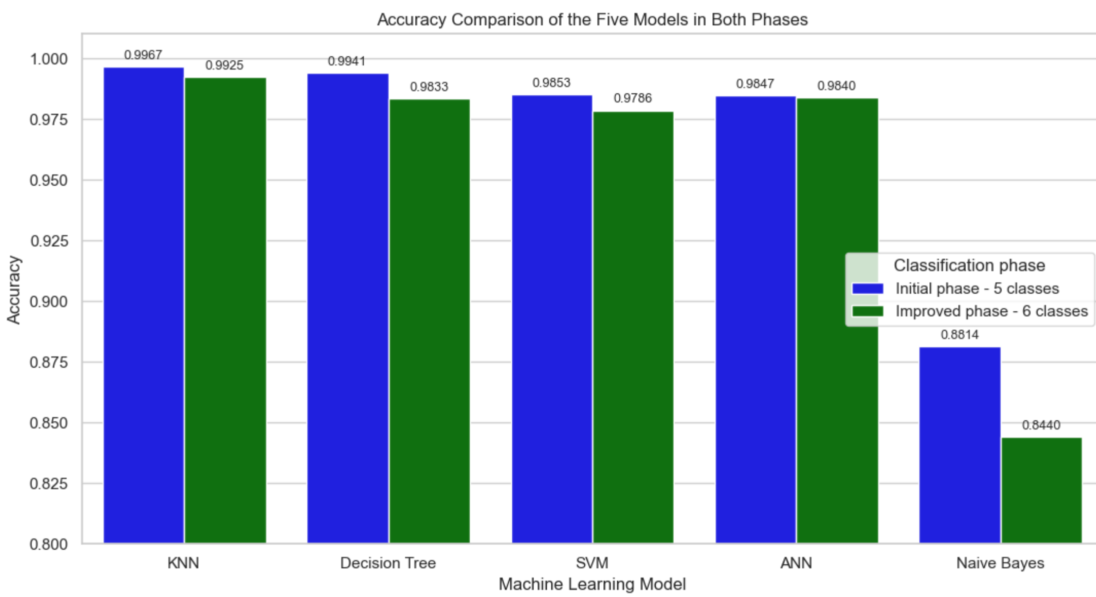

Figure 11: Accuracy comparison of the five machine learning models in the initial and improved classification phases.

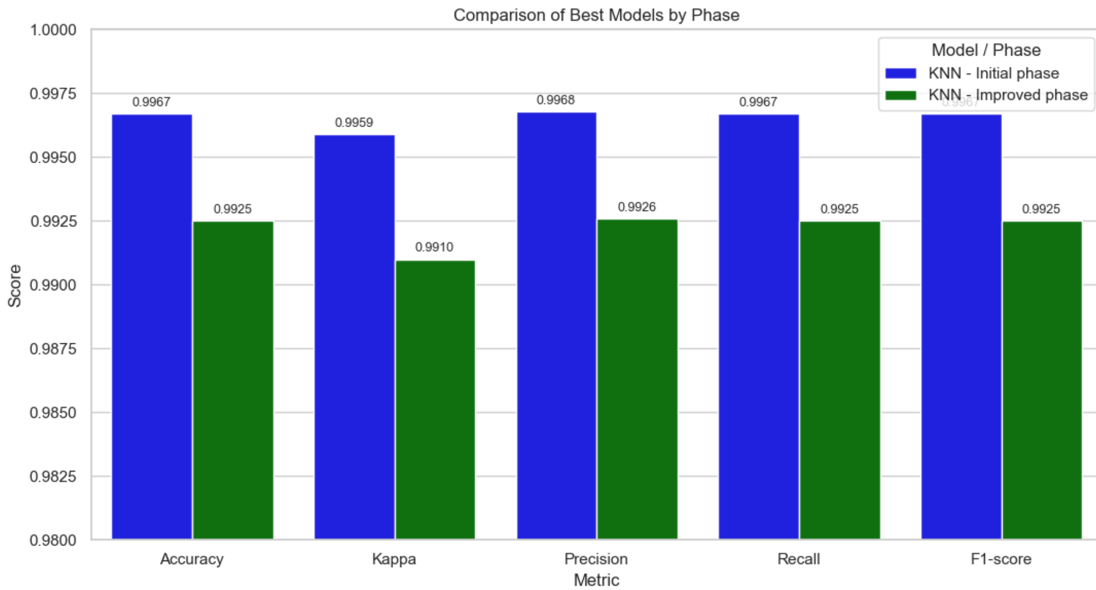

Figure 12: Comparison of the best-performing models in the initial and improved classification phases.

The cross-phase accuracy comparison shows that KNN achieved the highest accuracy in both workflows. In the initial five-class phase, KNN reached an accuracy of 0.9967, while in the improved six-class phase it reached 0.9925. Although the second workflow produced slightly lower accuracy values, all leading models remained highly competitive. This behavior is consistent with the increased complexity of the improved classification task, since the model had to distinguish six classes instead of five.

The best hyperparameters for the improved workflow were:

Table 20: Best hyperparameters obtained in the improved workflow.

Model	**Best hyperparameters**
Decision Tree	criterion = entropy; max_depth = 20; min_samples_split = 2
SVM	C = 10; gamma = scale; kernel = linear
ANN	activation = relu; hidden_layer_sizes = (50, 50); learning_rate_init = 0.01
KNN	metric = euclidean; n_neighbors = 3; weights = distance
Naive Bayes	var_smoothing = 1e-09

14 Improved KNN Classification Report

The improved KNN model obtained high performance across the six classes. The addition of Dark_ocean_water reduced the ambiguity between dark water and cloud shadow.

Table 21: Classification report for the improved KNN model.

Class	Precision	Recall	F1-score	Support
Bare_soil	1.00	1.00	1.00	980
Cloud	1.00	1.00	1.00	980
Cloud_shadow	0.99	1.00	1.00	980
Dark_ocean_water	0.99	0.97	0.98	980
Ocean_water	0.97	0.99	0.98	980
Vegetation	1.00	1.00	1.00	980
Macro average	0.99	0.99	0.99	5,880
Weighted average	0.99	0.99	0.99	5,880

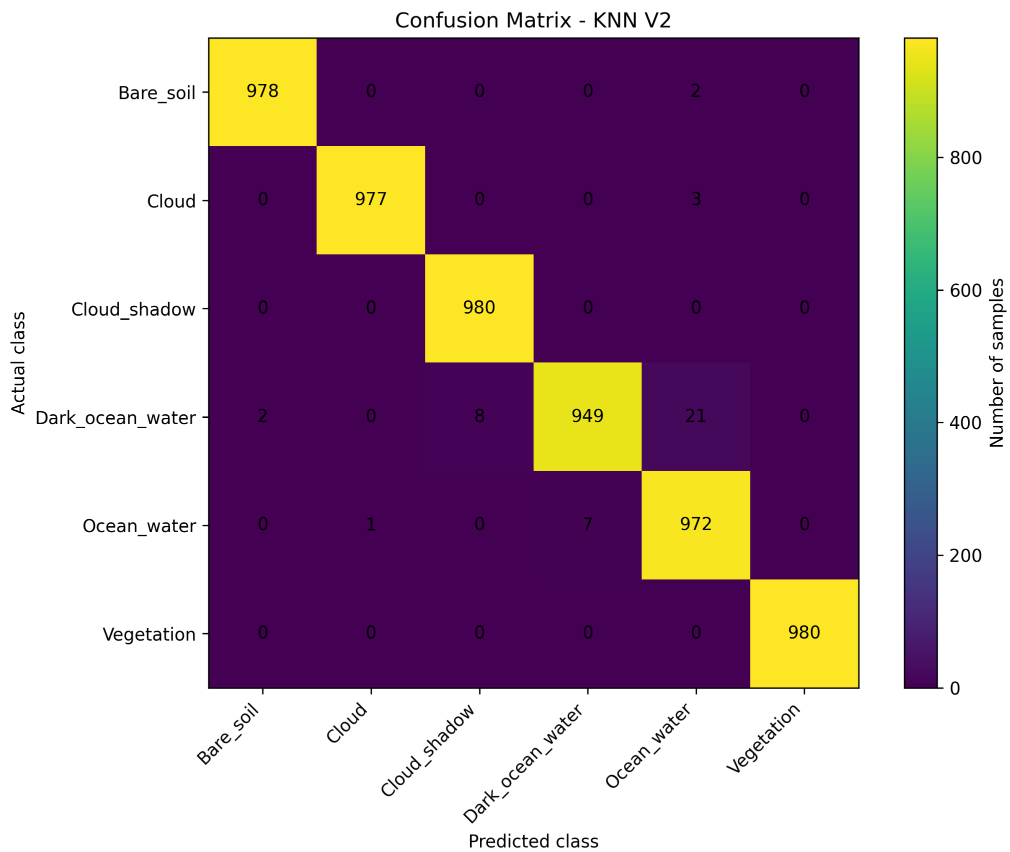

Figure 13: Confusion matrix for the improved KNN classifier.

15 Improved Raster Classification over the Cropped ROI

The improved KNN model was applied to the cropped region of interest. The output raster was saved as:

`classified_hunga_tonga_v2.tif`

The class codes used in the raster were:

Table 22: Raster class codes for the improved classification.

Raster code	Class name
1	Bare_soil
2	Cloud
3	Cloud_shadow
4	Dark_ocean_water
5	Ocean_water
6	Vegetation

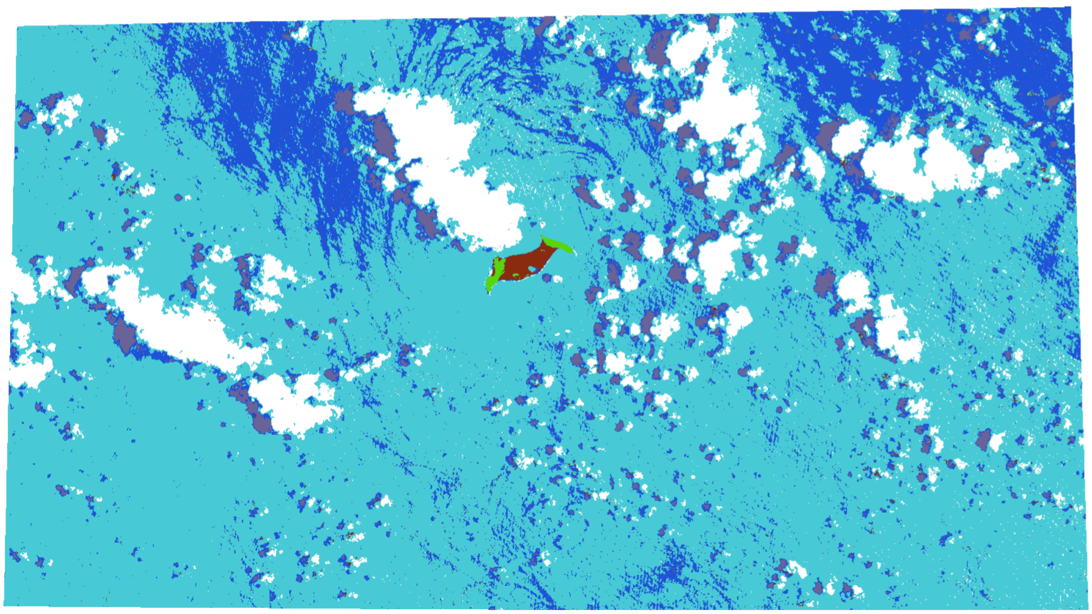

Figure 14: Improved classified map over the cropped region of interest.

Table 23: Area by class for the cropped ROI classified with the improved KNN model.

Class name	Pixel count	Area ha	Area km^2
Ocean_water	7,322,243	73,222.43	732.2243
Dark_ocean_water	1,496,831	14,968.31	149.6831
Cloud	1,022,928	10,229.28	102.2928
Cloud_shadow	283,709	2,837.09	28.3709
Bare_soil	54,330	543.30	5.4330
Vegetation	7,700	77.00	0.7700

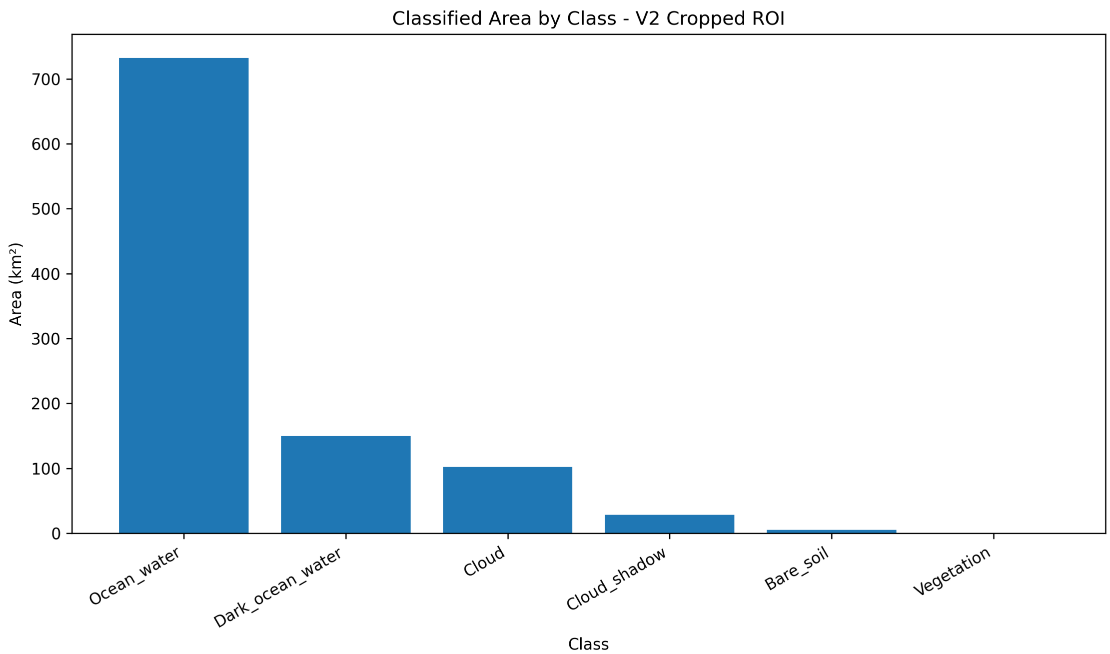

Figure 15: Area by class for the cropped ROI using the improved KNN model.

16 Full-Scene Classification Using the Improved Model

After validating the improved model over the cropped region, the same six-class KNN model was applied to the complete Sentinel-2 scene. This allowed the project to compare the local cropped classification with the full-scene generalization of the improved model.

The output raster was saved as:

`classified_hunga_tonga_full_scene_v2_model.tif`

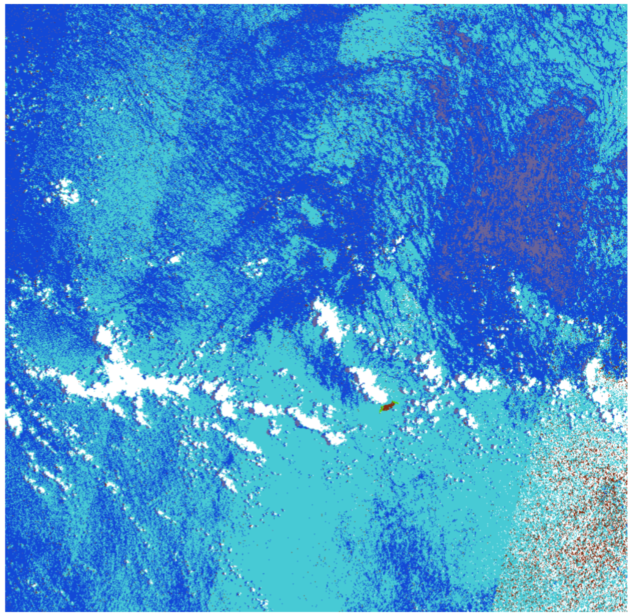

Figure 16: Complete Sentinel-2 scene classified using the improved six-class KNN model.

Table 24: Area by class for the complete scene classified with the improved KNN model.

Class name	**Pixel count**	**Area ha**	**Area km^2**
Ocean_water	57,029,761	570,297.61	5,702.9761
Dark_ocean_water	49,658,943	496,589.43	4,965.8943
Cloud	7,069,777	70,697.77	706.9777
Cloud_shadow	4,764,831	47,648.31	476.4831
Bare_soil	2,028,397	20,283.97	202.8397
Vegetation	8,686	86.86	0.8686

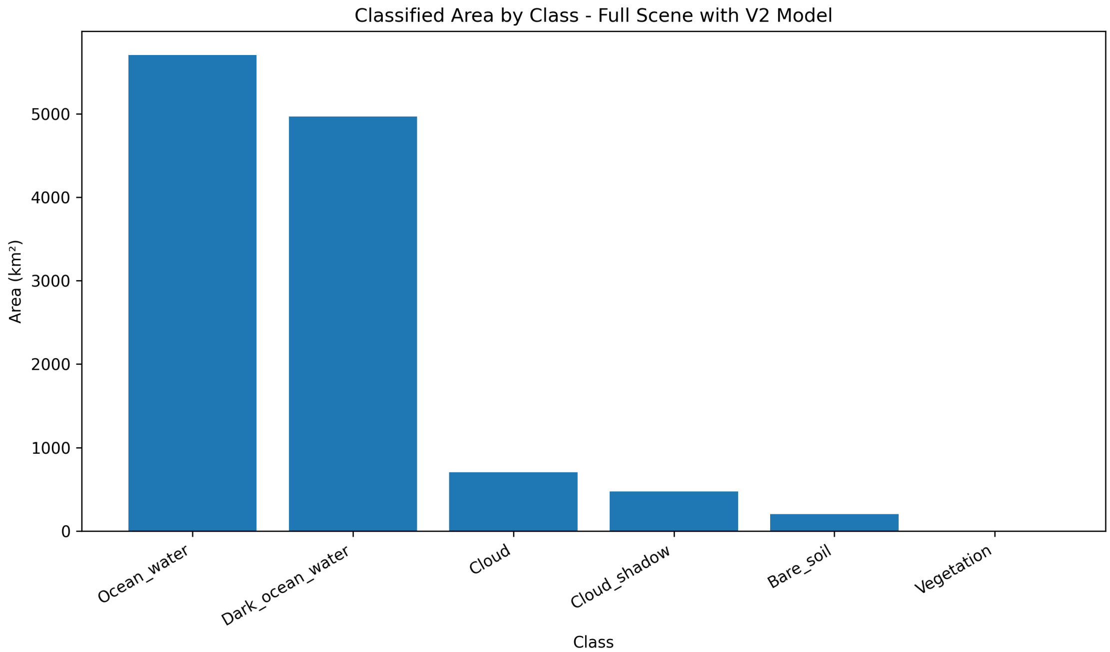

Figure 17: Area by class for the complete Sentinel-2 scene using the improved model.

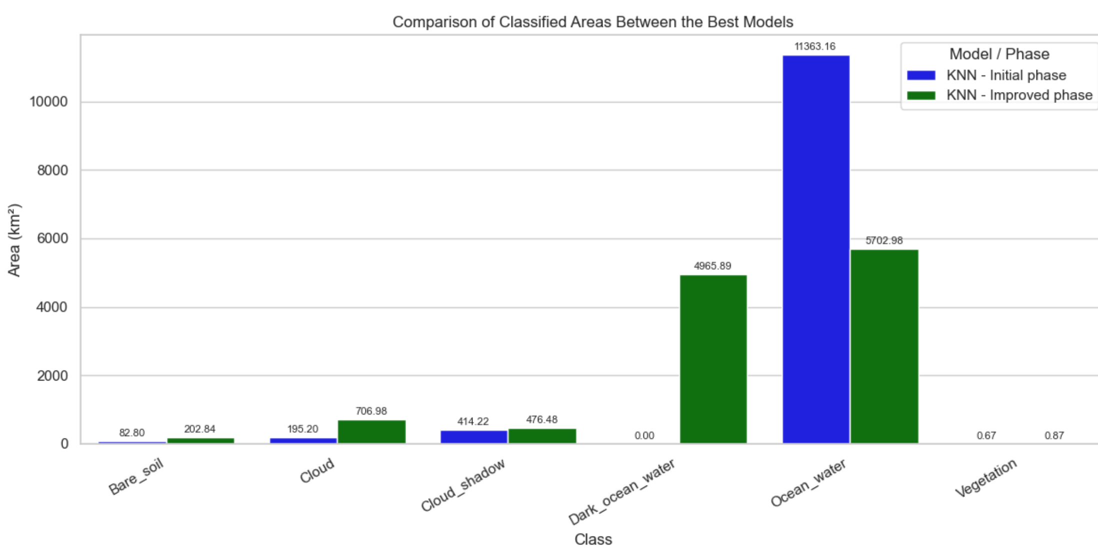

Figure 18: Comparison of classified areas between the best models of the initial and improved phases.

The classified area comparison shows that the improved model redistributed a significant portion of the original Ocean_water class into the new Dark_ocean_water class. In the initial phase, all ocean pixels were grouped into a single water category, while in the improved phase the model distinguished between regular ocean water and darker low-reflectance ocean zones. For this reason, the second model provides a more detailed thematic interpretation of the marine environment, even though its numerical metrics were slightly lower.

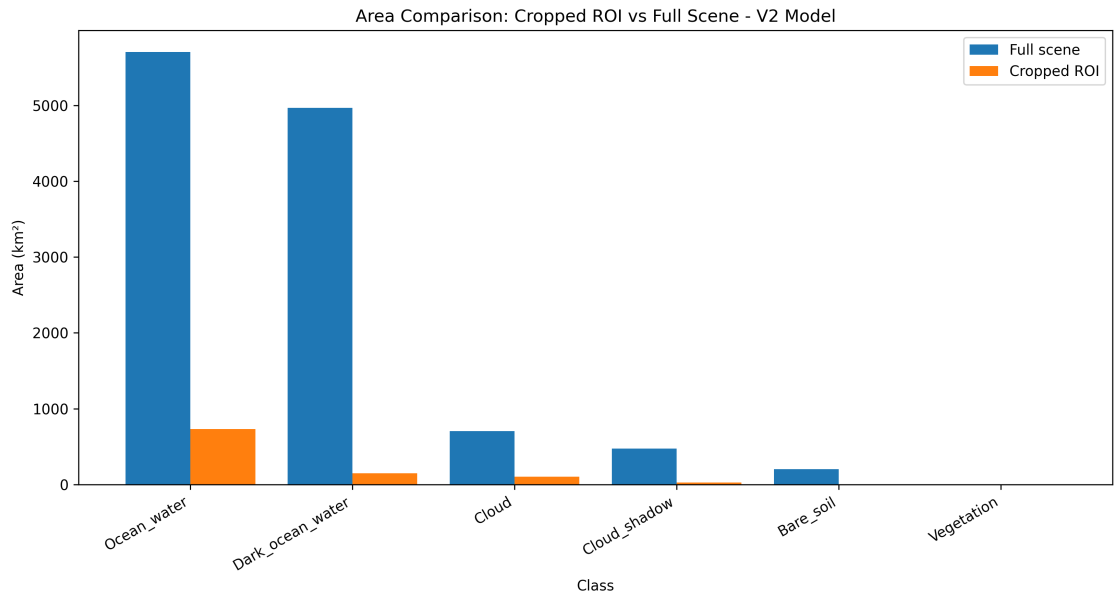

Figure 19: Area comparison between the cropped ROI and the complete Sentinel-2 scene using the improved model.

17 Discussion

The initial workflow demonstrated that the selected Sentinel-2 bands were highly effective for separating the five original classes. KNN achieved the highest performance, with an accuracy of 0.9967, a Kappa coefficient of 0.9959, and a macro F1-score of 0.9967. However, the confusion matrix revealed that some errors occurred between Ocean_water and Cloud_shadow. This was spectrally reasonable because both categories may present low reflectance values in optical bands.

The improved workflow addressed this limitation by introducing the Dark_ocean_water class. This class captured dark ocean pixels that were not necessarily cloud shadows. As a result, the six-class classification became more interpretable from a remote sensing perspective. Although the improved KNN model obtained slightly lower metrics than the initial KNN model, the decrease was small. The initial KNN model reached an accuracy of 0.9967, while the improved KNN model reached an accuracy of 0.9925. Therefore, both selected models achieved approximately 99% accuracy.

The slight reduction in the performance of the improved model can be explained by several factors. First, the classification problem became more complex because the number of classes increased from five to six. Second, the improved dataset contained more extracted training information: the initial dataset had 131,840 extracted samples, while the improved dataset had 265,651 samples, representing an increase of 133,811 samples, or approximately 101.49%. After balancing, the initial training dataset contained 12,280 samples, while the improved balanced dataset contained 23,520 samples, representing an increase of 11,240 samples, or approximately 91.53%. This larger dataset provided more information for model training, but it also introduced a more detailed classification structure.

In addition to the new class definition, the improved workflow also used a larger balanced dataset. The initial phase used 12,280 balanced samples, while the improved phase used 23,520 balanced samples. This represents an increase of 11,240 samples, equivalent to approximately 91.53%. Therefore, although the second model solved a more complex six-class problem, it also benefited from a larger training and testing base. However, the improved samples were extracted from a smaller cropped region of interest, which means that the model may be better adapted to the spectral conditions around the island. For this reason, the full-scene classification using the improved model should be interpreted as a spatial generalization test rather than as a completely independent validation over the entire scene.

The improved workflow also benefited from better polygon selection. The second set of polygons was created after reviewing the limitations of the first classification, which allowed the training samples to better represent the spectral differences between normal ocean water, dark ocean water, and cloud shadows. However, because the improved polygons were selected inside a smaller cropped region of interest, the model may contain some spatial bias. For this reason, the full-scene classification using the improved model should be interpreted as a spatial generalization experiment within the same Sentinel-2 image.

Even with this limitation, the improved model produced a more useful and detailed map than the first model. The first model grouped all ocean pixels into a single Ocean_water class, while the improved model separated Ocean_water and Dark_ocean_water. This distinction improved the thematic interpretation of the marine environment and reduced the visual ambiguity between dark water areas and cloud shadows.

Naive Bayes consistently obtained the lowest performance in both workflows. This result is likely related to its independence assumption, since Sentinel-2 spectral bands are often correlated. In contrast, KNN performed very well because spectrally similar pixels tended to cluster consistently in the feature space.

18 Final Outputs

The final project generated the following main outputs:

- Initial training dataset: `training_data.tsv`.
- Improved training dataset: `training_data_v2.tsv`.
- Trained models for Decision Tree, SVM, ANN, KNN, and Naive Bayes.
- Initial classified raster: `classified_hunga_tonga.tif`.
- Improved cropped raster: `classified_hunga_tonga_v2.tif`.
- Full-scene raster classified with the improved model: `classified_hunga_tonga_full_scene_v2_model.tif`.
- Confusion matrices and model comparison tables.
- Area quantification tables by class.
- QGIS map visualizations with discrete class symbology.

19 Conclusions

This project successfully implemented a complete supervised classification workflow for Sentinel-2 imagery over the Hunga Tonga–Hunga Ha'apai volcanic system. The workflow included preprocessing, band alignment, training polygon digitization, pixel extraction, dataset construction, model training, hyperparameter optimization, raster inference, QGIS visualization, and area quantification.

The five required machine learning architectures were implemented and compared: Decision Tree, Support Vector Machine, Artificial Neural Network, K-Nearest Neighbor, and Naive Bayes. In both the initial and improved workflows, KNN achieved the best overall performance. In the initial five-class workflow, KNN reached an accuracy of 0.9967, a Kappa coefficient of 0.9959, and a macro F1-score of 0.9967. In the improved six-class workflow, KNN reached an accuracy of 0.9925, a Kappa coefficient of 0.9910, and a macro F1-score of 0.9925.

The addition of the Dark_ocean_water class improved the thematic interpretation of the classification by separating low-reflectance ocean areas from cloud shadows. Although the improved model obtained slightly lower metrics than the initial model, both selected models achieved approximately 99% accuracy. The small reduction in performance is reasonable because the improved model solved a more complex classification problem with six classes instead of five. More importantly, the improved model generated a more detailed and useful thematic map, especially in the cropped region of interest, where the final classification showed strong visual agreement with the original Sentinel-2 image.

The final full-scene classification using the improved model demonstrated that the six-class model could be transferred to the complete Sentinel-2 scene. However, because the improved model was trained with samples from a smaller cropped ROI, the full-scene result should be interpreted as a spatial generalization experiment within the same image. This is an important methodological consideration, since the ROI-based training process may introduce bias toward the spectral conditions present around the island.

Overall, the project demonstrates that Sentinel-2 multispectral data combined with supervised machine learning can produce accurate and interpretable thematic classifications for volcanic island environments and surrounding ocean surfaces.

20 References

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